Entering the Car Docker

1. Course Content

- This lesson is only for users with **Raspberry Pi 5** and **Jetson-Nano** motherboards.

> [!NOTE]
>
> - Because the Raspberry Pi 5 and Jetson Nano system versions cannot directly install the `ros2 humble` environment, the `ros2 humble` environment is placed in `Docker`, and the host machine does not have a ROS2 environment.
>
> - After entering Docker with the Jetson Nano version, you need to open an rviz2 or rqt window before running the AI large-scale multimodal vision tutorial.

2. Basic Operations

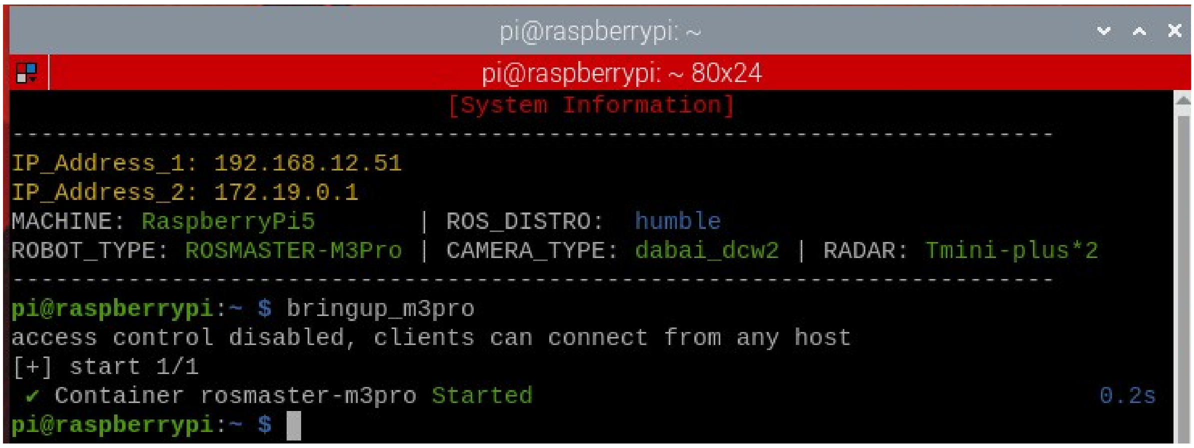
2.1 Launching the m3pro Container

- If you wish to run the example programs featured in this tutorial, you must first launch the `m3pro` Docker container.

> [!WARNING]
>
> - **Important:** Ensure that you launch the `m3pro` Docker container *within* the VNC graphical interface; otherwise, visualization windows (such as RViz) may fail to display correctly.
> - Regarding external peripherals—such as game controllers, AI voice modules, and other hardware devices—the Docker container will only load devices that were present at the moment the container was launched. If a hardware device is plugged in *after* the container has already started, it will not be automatically synchronized in real-time within the container. To recognize newly connected devices, you must stop and then restart the container.

```
bringup_m3pro
```

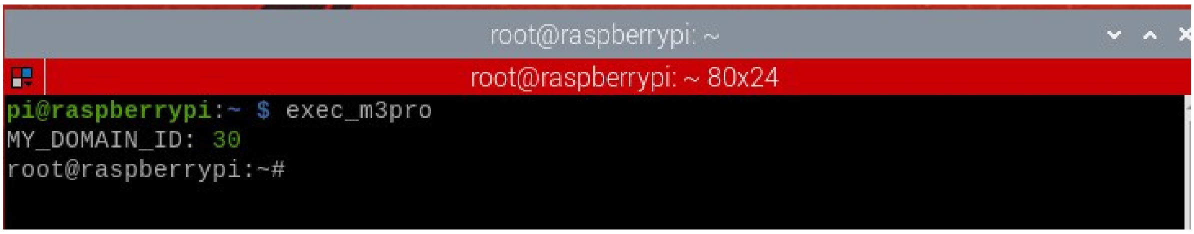
The message `container rosmaster-m3pro Started` indicates that the launch was successful.



2.2 Accessing the m3pro Container Terminal

- The commands featured in the subsequent tutorials must be executed *inside* the container. To open the container terminal, use the following command:

```
exec_m3pro
```

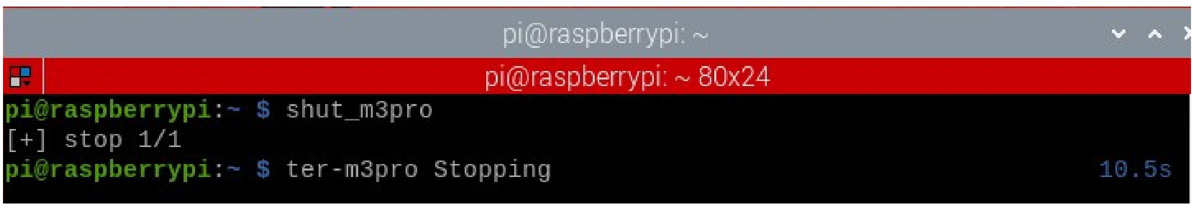


- To exit the container terminal, press `Ctrl + D`.

> [!IMPORTANT]
>
> **Important**
>
> - Unless explicitly stated otherwise, users of the Raspberry Pi and Jetson Nano platforms should assume that all subsequent program execution commands are to be performed *within* the **m3pro** container. You must first launch the container and then enter the commands within the container terminal. This requirement will not be reiterated in subsequent lessons. ### 2.3 Shutting Down the m3pro Container

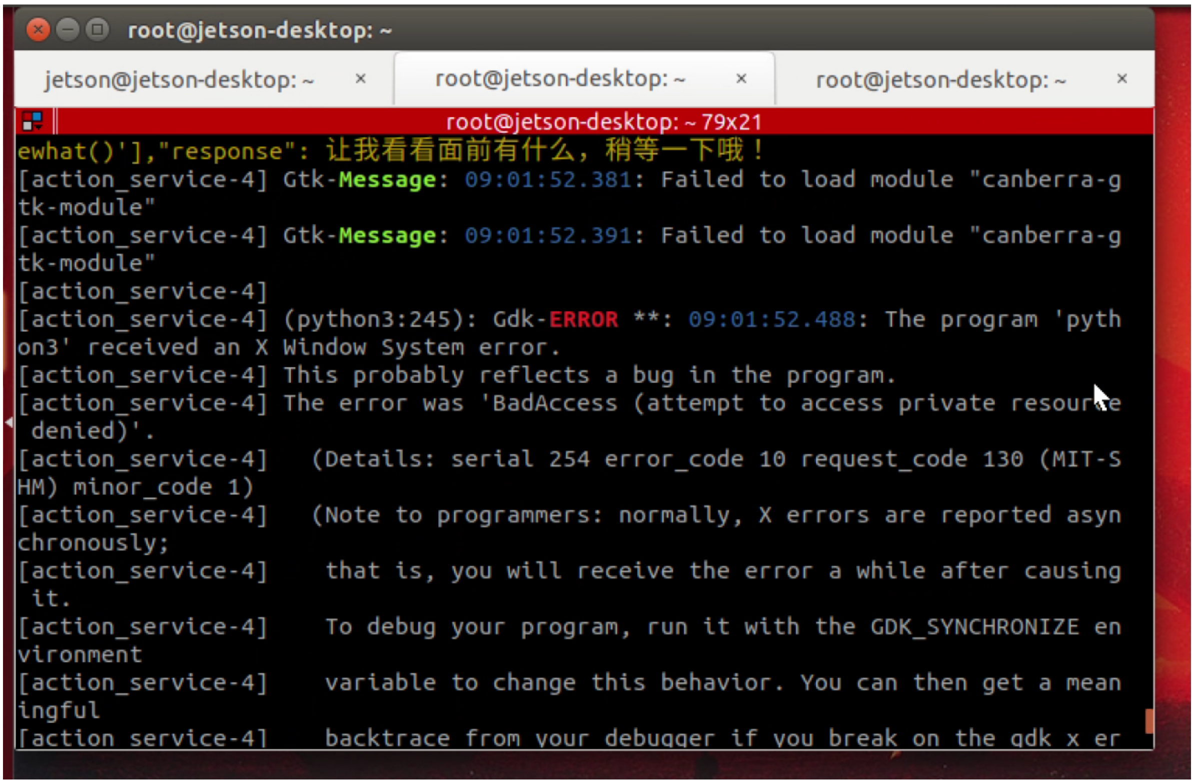
- If you genuinely no longer require the m3pro container and need to shut it down manually (typically, manual shutdown is not necessary):

```
shut_m3pro
```



2.4 Issues Encountered After Entering the Docker Container on Jetson Nano Versions of M3Pro, A1, M1, and M3 Robots

- On Jetson Nano versions, attempting to run the AI Large Model Vision examples immediately after entering the Docker container results in a "Permission Denied" error, as shown in the figure below.



- Solution: First, open a window within the Docker container using commands such as `rviz2` or `rqt`. After opening and then closing this window, you may proceed to run the AI Large Model Vision examples, as illustrated in the figure below.